RESEARCH ARTICLE

Does Food Processing Change Blood Sugar and Insulin Responses to a Calorie-Matched Lunch?

Abstract

Nutrition labels treat a calorie as a calorie, but the body may not respond to all calories the same way. We tested whether two lunches matched for labeled calories and macronutrients, but differing in processing level, produced different metabolic and later-eating responses. In a randomized within-subjects crossover study, 48 adults wore continuous glucose monitors and ate two 700-kcal lunches on separate days: a minimally processed lunch (rice, chicken or tofu, beans, vegetables, fruit, plain yogurt) and an ultra-processed lunch (nuggets, refined bun, sweetened yogurt drink, packaged snack bar). Blood draws before lunch and 60 minutes afterward measured insulin and triglycerides, and participants later had free access to a snack tray. The ultra-processed lunch produced a sharper glucose spike, a larger insulin rise, and a later dip below pre-lunch glucose levels, while the minimally processed lunch produced a flatter, more sustained glucose curve. Participants also ate about 140 more kilocalories from the snack tray after the ultra-processed lunch, and larger glucose drops predicted greater snack intake. Because this was a short, single-meal lab study, it cannot show how these patterns unfold over weeks or months. Still, it suggests that two meals with the same calorie label can produce different metabolic responses, which may shape how much people eat a few hours later.

Significance: Calorie labels assume that 700 calories from any food are metabolically interchangeable. By matching two lunches on labeled calories and macronutrients while varying processing level, this study tests whether processing is associated with differences in blood sugar, insulin response, and later eating — a question relevant to anyone packing a lunch, designing a cafeteria menu, or managing their own blood sugar.

Key Terms

Continuous glucose monitor (CGM): A wearable sensor that repeatedly estimates glucose levels over time.
Fasting baseline: A participant's glucose level before eating the lunch.
Insulin: A hormone involved in moving glucose from the blood into cells.
Glucose dip: A decline in glucose after its post-meal rise, sometimes below the pre-meal baseline.
Processing level: The extent to which a food has been altered from its original ingredients through industrial preparation.

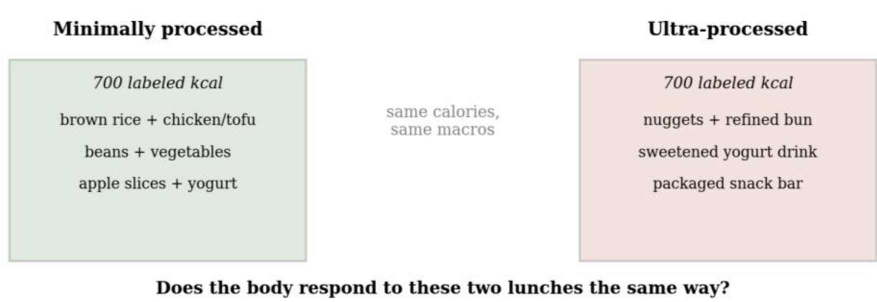

Figure 1. *Two lunches matched for labeled calories and macronutrients, differing only in processing level.*

Introduction

A calorie listed on a nutrition label is meant to be a fixed unit: 700 calories of one food should carry the same energy value as 700 calories of another. But the calorie count on a label comes from a fixed formula (Atwater & Bryant, 1900) that does not account for how processed the food is, or for how quickly that food moves through the stomach and gut. Cooking and milling break down plant cell walls and starch granules before food ever reaches the gut, and industrial processing goes further still, pre-digesting much of the work the body would otherwise do itself (Carmody & Wrangham, 2009). Foods that are easier to break down tend to leave the stomach faster, producing a sharper rise in blood sugar and a larger insulin response; that sharp rise is often followed by a steeper drop, sometimes below the level a person started at, which is linked to feeling hungry again sooner (Holt et al., 1995; Ludwig, 2002). If processing changes how quickly a meal raises and then lowers blood sugar, two meals with identical calorie labels could leave the body in very different states a few hours later.

Prior evidence points this way but has not isolated processing on its own. Hall et al. (2019) found that adults given an ultra-processed diet ate about 500 more calories per day than when given a minimally processed diet matched for calories, sugar, fat, fiber, and macronutrients, but that study measured a full diet over two weeks, not a single matched meal, so it could not isolate the effect of any one meal on fullness afterward.

This study asks a more everyday version of the question. Most people are not choosing between elaborate diet plans; they are choosing what to eat for lunch. We created two plausible cafeteria or meal-prep lunches, matched them on labeled calories and macronutrients, and varied only processing level. We then asked whether the two lunches produced the same blood-sugar response and whether people ate the same amount from an afternoon snack tray afterward.

Method

Design and lunches. Forty-eight healthy adults (54% female, ages 19–55) completed a randomized within-subjects crossover study with two lab visits at least one week apart. At each visit, participants ate one of two 700-kcal lunches matched for protein (~35 g), fat (~25 g), and carbohydrate (~80 g). The minimally processed lunch included brown rice, grilled chicken or tofu, black beans, mixed vegetables, apple slices, and plain yogurt. The ultra-processed lunch included chicken nuggets, a refined-flour bun or crackers, a sweetened yogurt drink, and a packaged snack bar. Lunches were classified using the NOVA system, and order was randomized and counterbalanced.

Measures. Participants wore a continuous glucose monitor (CGM), a small sensor patch on the back of the upper arm, from the morning of each visit, recording interstitial glucose every 15 minutes throughout the four-hour observation window. A venous blood sample was drawn once immediately before lunch (fasting baseline) and once 60 minutes after lunch, when insulin and triglyceride responses to a meal typically peak, and assayed for insulin and triglycerides; this kept blood draws to two per visit rather than a repeated-sampling protocol. At the four-hour mark, a snack tray (mixed sweet and savory items, weighed before and after) was made available, and participants could eat freely for 30 minutes; intake was converted to kilocalories from item weights. Glucose curves were summarized as peak value and minutes spent below fasting baseline; insulin, triglycerides, and snack intake were compared between conditions with paired t-tests, and snack intake was correlated with each person's lowest post-lunch glucose value.

Results

Blood sugar rose and fell more sharply after the ultra-processed lunch. Glucose increased at a similar rate for both lunches during the first 30 minutes, but the ultra-processed lunch then produced a higher peak (142 vs. 116 mg/dL) and a later dip below pre-lunch baseline. Over the next two hours, glucose spent 38 minutes below baseline after the ultra-processed lunch, compared with 6 minutes after the minimally processed lunch ($p < .001$; Fig. 2).

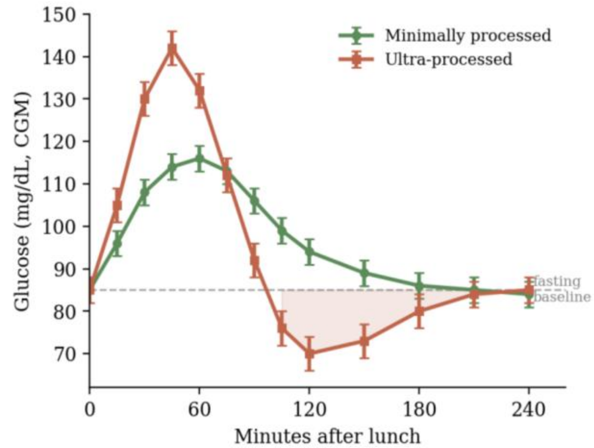

Figure 2. *Mean CGM-measured glucose over four hours after each lunch, relative to fasting baseline (dashed line). Shaded region shows time spent below baseline after the ultra-processed lunch. Error bars are ±1 SE.*

Insulin rose more after the ultra-processed lunch. At the 60-minute draw, insulin was higher after the ultra-processed lunch (52 µIU/mL) than after the minimally processed lunch (31 µIU/mL, paired $t(47) = 6.1$, $p < .001$); triglycerides were also modestly higher (118 vs. 104 mg/dL, paired $t(47) = 2.4$, $p = .02$), though this rests on a single post-meal timepoint and should be read as a smaller, more tentative finding than the glucose and insulin results (Table 1).

Measure	Min. processed	Ultra-processed
Glucose peak (mg/dL)	116	142
Time below baseline (min)	6	38
Insulin at 60 min (µIU/mL)	31	52
Triglycerides at 60 min (mg/dL)	104	118
Afternoon snack (kcal)	186	329

Table 1. *Summary of the main outcomes by lunch condition. Both lunches were matched at 700 labeled kilocalories.*

People ate more later after the ultra-processed lunch, in proportion to how far their blood sugar had dropped. At the afternoon snack tray, participants ate an average of 329 kcal after the ultra-processed lunch versus 186 kcal after the minimally processed lunch — about 140 more kilocalories, roughly a fifth of the lunch itself (paired $t(47) = 5.9$, $p < .001$; Fig. 3). The two effects were related: participants whose glucose fell further below baseline also tended to eat more at the snack tray ($r = .44$, $p = .002$), consistent with the glucose dip itself contributing to the drive to eat again.

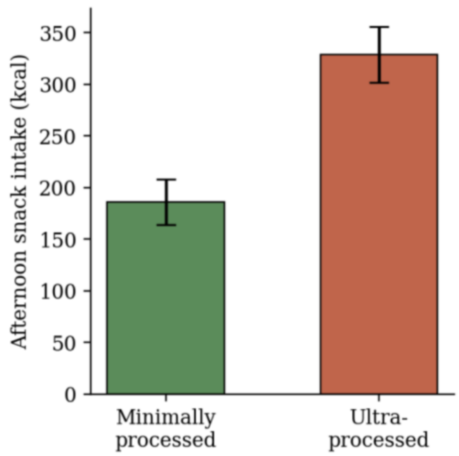

Figure 3. *Mean afternoon snack intake (kcal) by lunch condition. Error bars are ±1 SE.*

Conclusion

Although the two lunches were matched on labeled calories and macronutrients, they produced different metabolic and later-eating responses. The ultra-processed lunch was followed by a higher glucose peak, a larger insulin increase, more time below baseline glucose, and greater afternoon snack intake than the minimally processed lunch. Participants whose glucose fell further also tended to consume more from the snack tray. These findings indicate that meals with the same labeled calorie content can be associated with meaningfully different short-term physiological responses and subsequent eating behavior.

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